

COMP 333 — End-to-End Data Analytics Pipeline

Executive Summary

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1. Project Overview

This project implements a complete, end-to-end data analytics and machine learning pipeline applied to **New York City (NYC) Yellow Taxi Trip Records** for June–July 2025. **Phase 1** covers data acquisition, wrangling, EDA, and baseline modelling; **Phase 2** advances into feature engineering, supervised classification, unsupervised clustering, and result interpretation.

Attribute	Detail
Primary Dataset	NYC Yellow Taxi Trip Records — June & July 2025 (~7.3 M rows, 1.4+ GB on disk)
Supplementary	NYC Hourly Weather (Open-Meteo API) — temperature (°C) and precipitation (mm)
Environment	Google Colab (12.7 GB RAM); Python 3 · pandas · scikit-learn · XGBoost

2. Phase 1 — Data Acquisition, Cleaning & Baseline Modelling

Data Wrangling & EDA Highlights

- Removed 4 exact duplicates and **150,444 mirrored records**; dropped **238,440 zero-distance rows**; imputed missing values and applied Winsorization (99th-percentile) to five financial features.
- Final clean dataset: **14 features** across ~7 M records (Parquet). Trip distance is the strongest fare predictor ($r = 0.87$); credit card accounts for ~75 % of rides; demand peaks post-17:00 on weekdays.
- Baseline (Phase 1):** Linear Regression Model B (*trip_distance* + *tip_amount*) selected for low MSE, high R^2 , and stable generalisation.

3. Research Questions

RQ1 — Supervised Learning	RQ2 — Unsupervised Learning
<i>"Can we classify a high-tipper passenger based on trip features (distance, time of day, day of week, rate code, passenger count) and weather conditions?"</i>	<i>"Can we identify distinct natural clusters of NYC taxi trips based on temporal and financial features to study customer behaviour?"</i>
Target: is_high_tip ($\text{tip} \geq 20\%$ of <i>fare_amount</i>) — Models: Random Forest, XGBoost	Method: PCA + MiniBatch K-Means ($k = 7$)

4. Phase 2 — Feature Engineering

Thirty features were engineered across three groups: **(1)** log-transforms and cyclical hour/day encodings; **(2)** domain flags (*is_rush_hour*, *is_airport*, *is_credit_card*, *fare_per_mile*); **(3)** polynomial/interaction terms (distance^2 , $\text{distance} \times \text{passengers}$, $\text{temp} \times \text{precipitation}$). Filter, embedded (RF importance), and wrapper (RFE) methods produced a **top-12 composite feature set (FINAL_FEATURES)**, excluding leakage variables (*tip_amount*, *total_amount*, *tip_ratio*).

5. Supervised Learning Results (RQ1)

Both models trained on **1 M stratified credit-card records** (70/15/15 split, RandomSearchCV 3-fold, F1 metric). **XGBoost selected** for marginally higher AUC and 40 % faster training.

Metric	Random Forest	XGBoost
CV / Val F1-Score	0.8637 / 0.8636	0.8637 / 0.8636
Test Accuracy	0.77	0.77
Recall — High Tip	1.00	1.00
AUC-ROC	0.6293	0.6302

Training Time	~250 s	~150 s
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- Top features: **improvement_surcharge**, **fare_amount**, **congestion_surcharge** — tipping is driven by payment method and fare structure, *not* time of day or weather.
- Limitation: persistent 75/25 class imbalance causes low recall (0.08) for 'Low Tip'; cash rides introduce structural label noise. Future: SMOTE or undersampling.

6. Unsupervised Learning Results (RQ2)

PCA reduced the 8-feature space to **6 PCs** ($\approx$ **90 % variance**); elbow method on a 100 K subsample identified **k = 7**. Full dataset (~7 M rows) clustered with **MiniBatch K-Means** (Silhouette = 0.18; Davies-Bouldin = 1.35 — overlapping clusters consistent with real-world behavioural data).

Cluster	Label	Key Characteristic
0	Long Trip	Highest mean trip distance — airport or inter-borough transfers
1, 2	Short Trip	Low distance; likely artificial split of one natural group
3	Medium Trip	Mid-range distances; no strongly differentiating secondary feature
4	Weather-Impacted	Very high mean precipitation — rainy-day demand spike
5	Peak Morning	Lowest mean pickup hour — morning commute segment
6	High-Occupancy	Highest mean passenger count — group trips

7. Conclusions & Actionable Insights

- **RQ1 — Tipping Behaviour:** XGBoost classifies high-tippers with 77 % accuracy (AUC 0.63). Tipping is driven by **payment method and fare economics** — not weather or time of day. Drivers can expect higher tips on credit-card, high-fare, or airport-rate trips.
- **RQ2 — Trip Archetypes:** Seven clusters reveal a service dominated by **short and medium city trips**, a strong **morning-commute segment**, minority long-distance/airport transfers, and a distinct rainy-day cluster — actionable for targeted pricing and dispatching strategies.
- **Pipeline scalability:** Processing 1.4+ GB within Colab's 12.7 GB RAM required aggressive memory management (GC, dtype downcasting, stratified sampling); the pipeline is transferable to cloud-scale environments.